

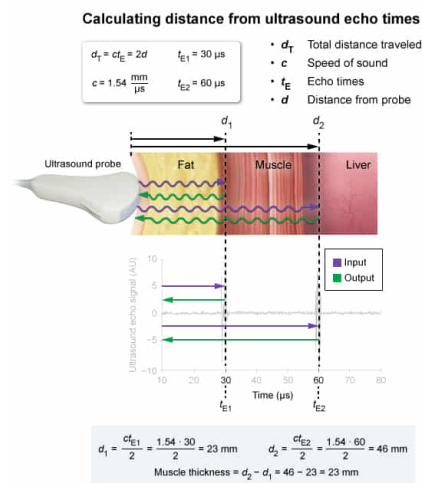
An ultrasound probe recorded the echo data shown in the figure above, representing the time required for the ultrasound signal to travel through the body and reflect back to the probe. The ultrasound signal passed through subcutaneous fat, muscle, and liver, resulting in echoes at 30 μs and 60 μs from the tissue interfaces. What is the thickness of the muscle tissue that the ultrasound wave traveled through? (Note: Use 1.54 mm/ μs for the speed of sound in the body.)

- ✓ ☒ A. 23 mm [13%]
☐ B. 30 mm [9%]
☐ C. 46 mm [73%]
☐ D. 60 mm [3%]

Correct

12% answered correctly

Explanation:



The total **distance** d_T that the ultrasound signal travels is calculated by multiplying the **speed of sound** c by the **echo time** t_E :

$$d_T = ct_E$$

The ultrasound signal travels from the ultrasound probe to the tissue interface and back to the probe again. Therefore, the distance d from the probe to the tissue interface is half the total distance traveled:

$$d = \frac{d_T}{2}$$

$$d = \frac{ct_E}{2}$$

In this question, the first echo is received 30 μs after the pulse is transmitted into the body. This echo is generated from the ultrasound signal reflecting off the interface between the fat and muscle tissues.

Hence, the distance from the probe to the fat/muscle interface d_1 is calculated as:

$$d_1 = \frac{1.54 \frac{\text{mm}}{\mu\text{s}} \times 30 \mu\text{s}}{2} = 23 \text{ mm}$$

Similarly, the distance from the probe to the muscle/liver interface d_2 depends on the second echo time (60 μs) and is calculated as:

$$d_2 = \frac{1.54 \frac{\text{mm}}{\mu\text{s}} \times 60 \mu\text{s}}{2} = 46 \text{ mm}$$

Therefore, the thickness T_M of the muscle layer is the difference of these two distances:

$$T_M = d_2 - d_1 = 46 \text{ mm} - 23 \text{ mm} = 23 \text{ mm}$$

(Choice B) The first echo is received by the probe at 30 μs , but the echo time must be multiplied by the speed of sound to calculate the total distance traveled.

(Choice C) 46 mm is twice the muscle thickness. The distance between the probe and the tissue interfaces depends on half the time for the ultrasound signal to be reflected back to the probe.

(Choice D) The second echo is received by the probe at 60 μs , but the echo time must be multiplied by the speed of sound to calculate the total distance traveled.

Educational objective:

Given the speed of sound in the body, the time required for an ultrasound signal to be reflected from a tissue interface can be used to calculate distance. The distance from the probe to the tissue interface is half the total round-trip distance.

Time spent: 0 sec
Subject:
 Physics

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System:
 Light and Sound